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Fatigue of the Erector Spinae Muscles

A Quantitative Assessment Using "Frequency Banding" of the Surface Electromyography Signal

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Study Design. The authors investigated fatigue-induced changes in the frequency content of the surface electromyographic (EMG) signal from the erector spinae muscles.

Objectives. The objective of the study was to understand the EMG changes in fatiguing muscle and to obtain a reliable index of fatigue.

Summary of Background Data. Power spectral analysis has been used increasingly in recent years to monitor muscle fatigue, but parameters other than the mean or median frequency have received little attention.

Methods. Thirty-five healthy volunteers participated. They pulled upward with constant force on a handlebar attached to a floor-mounted load cell while the EMG signal from the erector spinae was recorded at the levels of T10 and L3 at 1024 Hz; 1.0-sec "windows" of the signal were analyzed using fast Fourier transforms, and the resulting power spectra were divided into 10 frequency bands between 5 Hz and 300 Hz. The median frequency, total power, and peak amplitude of the spectra were also calculated. Changes in the frequency content of the EMG signal were examined during submaximal contractions of different intensity and duration.

Results. Median frequency decreased steadily during the contractions, whereas total power and peak amplitude increased. The most repeatable and linear index of change was the increase in the EMG signal in the 5–30 Hz frequency band. The middle-to-high frequency component of the EMG signal increased during the early stages of the contractions, but decreased as the endurance limit was approached.

Conclusions. Changes in the 5–30 Hz band of the EMG power spectrum provide a more reliable and linear index of fatigue in the erector spinae muscles than do changes in median frequency. In the erector spinae, the early effects of fatigue appear to be delayed by the recruitment of additional motor units. [Key words: erector spinae, fatigue, EMG, power spectrum] **Spine 1995;20:149–159**

The back muscles normally act to protect the lumbar spine, particularly during forward bending movements[2] but their protective action may be impaired if they become fatigued. This possibility could be investigated if techniques were available for the accurate and objective assessment of back muscle fatigue. In recent years, surface electromyography (EMG) has been used to monitor fatigue in a variety of muscles,[4,8,12,31] but there are problems to consider when applying the same techniques to the erector spinae.

First, the erector spinae are unusual muscles in that the slow-twitch (Type I) fibers are larger in cross-section than the fast-twitch (Type II) fibers, both in patients with low back pain[5,41] and people without back pain.[35,41] The large fibers therefore may be recruited at lower forces than the smaller fibers, and this unusual pattern of recruitment may explain atypical changes in the frequency content of the surface EMG with increasing force of contraction.[37] Another problem is that the erector spinae consist of a large number of discrete muscle fascicles that are innervated separately and can contract independently.[10,30] The relative activity of thoracic and lumbar fascicles depends on posture[3,17,19] and may also be influenced by the level of force and fatigue. The thoracic portion of the erector spinae has a larger percentage cross-sectional area of Type I fibers than the lumbar musculature,[41] and this may reflect a functional difference in the two regions, with the activity of the thoracic muscles being more tonic in character than that of the lumbar muscles, which are relatively inactive during standing postures.[24] Another unusual property of the erector spinae is that they rarely shorten beyond their length in the upright standing position; in fact, they perform most of their work when extending the upper body from flexed postures in which the muscles are lengthened considerably. All of these factors may cause the erector spinae to show atypical EMG changes during sustained fatiguing contractions.

A further problem with assessing erector spinae fatigue from surface EMG is that the EMG techniques themselves have given conflicting results. Both the root mean square (RMS) and integrated EMG signals generally increase during sustained isometric contractions,[14,20,36,43,44] but under certain circumstances, they may remain constant or even decrease.[13,39] An improvement to using the RMS or integrated EMG signal is to analyze the raw EMG signal using fast Fourier transforms to

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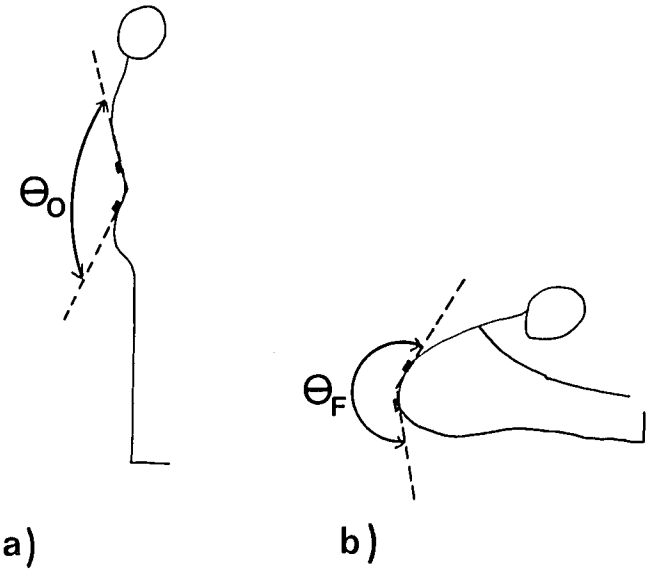


Figure 1. The Isotrak measures the lumbar curvature between L1 and S1 in the sagittal plane. The range of lumbar flexion is determined from changes in lumbar curvature between the erect standing (A) and fully flexed (B) positions and is given by $\Theta_O - \Theta_F$.

express its frequency content as a "power spectrum." As muscle fatigues, the power spectrum is shifted to lower frequencies, which are quantified by measuring the change in the mean or median frequency. Most studies report a decrease in mean and median frequency during constant force contractions,[25,36,37] but several have found that the low frequency shift is more pronounced in the early stages of a contraction.[6,8,39] This makes it difficult to obtain an unambiguous measure of "fatigue."

This study investigates, in detail, the fatigue-related changes in the surface EMG signal from the erector spinae muscles. Changes in discrete frequency bands of the power spectrum were compared with changes in several "power spectrum" parameters including the median frequency. In this way, we hoped to explain some of the previous inconsistencies regarding EMG measurements of fatigue. Also, we aimed to identify the EMG parameters that afforded the most repeatable and linear indices of fatigue under different loading conditions.

Materials and Methods

Subjects Participating in the Study. Fifteen men and twenty women without histories of serious low back pain participated. The average age was 30.7 years (SD 7.9, range: 19–43 years). Average body mass for the men was 78.4 kg (SD 12.1, range: 53.0–95.5 kg) and for women was 57.9 kg (SD 13.0, range: 48.0–94.1 kg).

Muscle Length and Spinal Posture. The length of the erector spinae muscles is largely determined by the curvature of the lumbar spine in the sagittal plane, so this was measured and controlled in all experiments, using the 3-Space Isotrak device (Polhemus). The Isotrak consists of a source and sensor of pulsed electromagnetic waves that can be attached to the skin surface overlying the spinous processes of L1 and S1: the angle between source and sensor (Figure 1A) is then a good measure of lumbar curvature, and this is sampled at 60 Hz and input to a microcomputer.[1] The range of lumbar flexion was established for each subject by measuring lumbar curvature in the erect standing (Figure 1A) and extreme toe-touching (Figure 1B) positions. This enabled a given posture, and hence muscle length, to be defined in terms of the full range of flexion of the lumbar spine, for example, "70% flexion."

Maximum Voluntary Contractions (MVCs). Each subject stood in a wooden frame with the front of the thighs resting against a cushioned crossbar and with a strap fastened around the hips. While leaning forward in the frame, each subject pulled upward on a handlebar attached by a chain to a floor-mounted load cell (Figure 2). The handlebar was constrained by the frame to move only in the vertical direction, and its height was adjusted so that the subject's lumbar spine was flexed by 70%, as measured by the Isotrak. The position of the feet and the lever arms D and d (Figure 2) were recorded as described[16] so that the posture could be reproduced exactly in subsequent tests. Subjects then pulled up with maximum force for 3.3 sec, while the output from the load cell was A-D

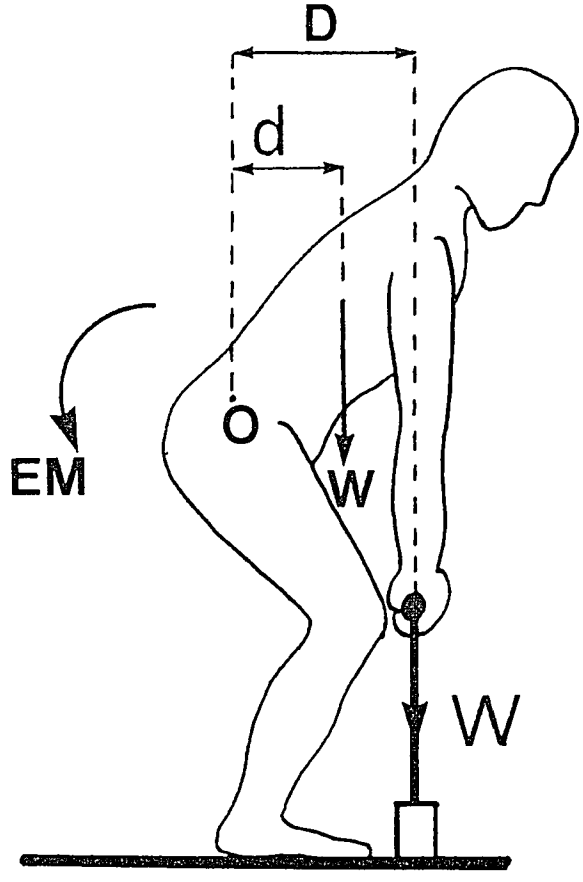


Figure 2. EMG activity of the erector spinae muscles was recorded while each subject pulled upward on a floor-mounted load cell with a constant force (W). The extensor moment EM generated by the back muscles was calculated from W, the upper body weight (w), and the lever arms D and d.

converted at 60 Hz and input to a microcomputer.[16] (The precise timing of this, and other tests, was chosen for convenience in data acquisition.) To avoid spurious high transient forces resulting from jerking movements, the force data was averaged over 10 discrete intervals of 0.33s, and the maximum of the 10 values was retained. The total extensor moment generated by the back muscles was calculated from the lever arms D and d (Figure 2), body mass (w), and the force on the load cell (W),[16] and this was used as a measure of the maximum force generated by the erector spinae. Between two and four repetitions were necessary to establish a consistent maximum. The Isotrak was used to check that subjects maintained the same posture during each pull.

Electromyography. The electrical activity of the erector spinae muscles was measured using disposable surface electrodes (Biolect, UK) attached over the muscle belly on the left-hand side at T10 and L3. The use of special skin preparation pads (Biolect, UK) ensured that skin resistance was always less than 10 kohms; often it was less than 2 kohms. The location of electrodes was chosen to obtain EMG signals representative of the main muscle mass of erector spinae at both lumbar and thoracic levels.[16] The raw EMG signal from each electrode was filtered with a band-pass of 5–300 Hz, amplified, A-D converted at 1024 Hz, and stored on a microcomputer for subsequent analysis.

Frequency Analysis of EMG Signal. Raw EMG data was analyzed in the frequency range 1–512 Hz using a fast Fourier transform algorithm acting on 1024 data points. These data were recorded over a period of 1.0 sec to give a resolution of 1 Hz. Power density spectra were constructed for frequencies in the restricted range 5–300 Hz because preliminary tests showed that most of the signal above 300 Hz was noise, and signal below 5 Hz was most susceptible to movement artifact. High frequency aliasing was avoided by sampling at 1024 Hz (twice the Nyquist frequency), and the accuracy of the fast Fourier analysis was checked over a range of frequencies using a signal generator. The following power spectral parameters, defined in Figure 3, were calculated: median frequency, peak amplitude, peak amplitude frequency, and total power. The electrical "power" was calculated in each of 10 frequency bands; the first band was 5–30 Hz; the remainder were 30–60 Hz, 60–90 Hz, and so on up to 300 Hz. The full analysis was repeated using the square root of "power" (i.e., the amplitude) at each frequency, to obtain an "amplitude spectrum." This was expected to be more sensitive to changes in the highest and lowest frequency components of the EMG signal (see below).

Fatiguing Isometric Contractions. Five minutes were allowed for recovery after the final MVC test. The subject then was repositioned exactly in the frame and pulled up on the handlebar for a period of 46 seconds, with a force that was calculated to generate 60% of his or her previously measured maximum extensor moment ("60% MVC"). Electromyographic activity was sampled for 1.0-sec periods, at the start of the pull and subsequently at 5.0-sec intervals. If the force differed by more than 5% from the target, then the test was discontinued.

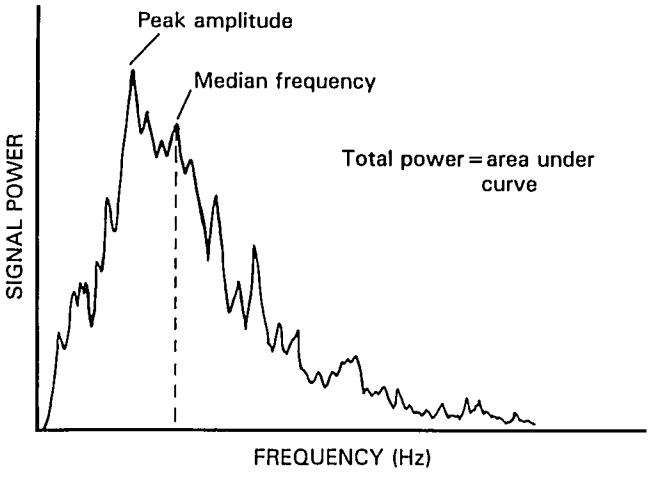


Figure 3. The frequency analysis of the EMG signal resulted in a "power spectrum," which is a histogram of signal intensity plotted against frequency. The total power is the area under the curve, and it represents the overall size of the EMG signal at all frequencies. The median frequency is that frequency that splits the area under the curve into two equal halves. The peak amplitude is the highest amplitude in the entire spectrum, and the frequency at which it occurs is the peak amplitude frequency.

Effect of Increasing Load and Duration. Nine subjects continued the 60% MVC fatiguing contraction "to endurance," that is, until they could no longer maintain the force above 95% of the target level. On a separate day, the same subjects also performed sustained contractions at 80% MVC for 23 sec, or to endurance, whichever was the shorter. In this case, the EMG signal was sampled for 1.0-sec periods, at the start of the pull and at 2.4-sec intervals thereafter (2.4 sec was the minimum time required to record and save the EMG data for each sampling period). These tests enabled us to examine the effects of increasing duration and of increasing contraction force on the EMG signal.

Assessment of Repeatability. Ten subjects repeated the submaximal test at 60% MVC approximately 1 hour after completing the initial test. The electrodes were not removed between the tests and so the within-day repeatability obtained from these measurements is an indication of the variability of the EMG signal when the test conditions remain the same. Six of the ten subjects also repeated the test several days later. In this case, the whole procedure was performed anew, including palpating the spine, attaching the electrodes and the 3-Space Isotrak, remeasuring the person's range of flexion, and remeasuring their MVC. The day-to-day repeatability estimates therefore include all the possible sources of variation that would be encountered if the test were to be used in longitudinal studies of subject performance.

Statistical Analysis. Most of the EMG changes during sustained contractions were approximately linear with time, so linear regression was used to calculate the initial value (the intercept) and its rate of change with time (the gradient). To standardize results between subjects, changes with time were evaluated from the regression line and expressed as a percent of the initial value. Differences between initial and final values

Table 1. Initial Values of Median Frequency and Total Power in Each of the Three Fatigue Tests

		Amplitude Spectrum			Power Spectrum		
	Site	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)
Median frequency (Hz)	T	68	68	69	53	53	56
	L	59	59	55	40	40	39
Total power (arbitrary units)	T	21,272	21,343	26,724	3.6×10^6	3.4×10^6	6.3×10^6
	L	20,303	20,347	26,239	3.4×10^6	2.9×10^6	7.4×10^6

Three fatigue tests are: 1) 60% maximum voluntary contraction (MVC) for 46 seconds (60% submax), 2) 60% MVC to the limit of endurance (60% LOE), and 3) 80% MVC for 23 seconds (80% submax). Data are shown for the amplitude and power spectra from EMG recordings made at the the level of the 10th thoracic (T) and third lumbar (L) vertebrae.

were compared using a matched-pair Student *t*-test. Associations between EMG parameters and endurance time were evaluated using linear regression to calculate the Pearson product correlation coefficient. Reliability estimates for the within-day and day-to-day repeated trials were evaluated using a one-way analysis of variance with repeated measures that allowed calculation of the intraclass correlation coefficient. Significance was measured at the 5% level.

Results

Frequency Spectrum Parameters

The values of median frequency and total power at the start of the three tests are compared in Table 1. The stronger muscle contractions required in the 80% MVC test increased total power compared with the 60% MVC tests, and this reached significance at L3. The initial median frequency was similar in all three tests.

The effect of a severe fatiguing contraction on the power spectrum is depicted in Figure 4; power at high frequencies decreased, and that at low frequencies increased. Individual subjects often showed unequal changes at T10 and L3, but there was no consistent tendency for the changes to be greater at one level than the other, and the averaged results (see below) showed no important differences between T10 and L3.

Changes in the spectral parameters during the three fatigue tests are summarized in Table 2. In all cases, the median frequency decreased and the peak amplitude and total power increased; all of these changes were significantly greater than zero. Changes in peak amplitude frequency were small and inconsistent. In subjects who performed all of the tests, a matched-pair comparison showed that the rates of change in the spectral parameters with respect to time (the gradients) were greater during the 80% submaximal test than the 60% submaximal test. These changes were significant for all parameters at L3 and for the median frequency at T10. However, there was no significant difference in the gradients between the 60% submaximal test and the 60% endurance test. This latter comparison was made in only seven of the nine subjects who performed the endurance tests because two subjects reached their endurance limits within the 46 sec of the submaximal test. However, it indicates that the changes in the power spectral parameters remained constant throughout the 60% fatiguing contraction.

The linearity of the changes with respect to time during the sustained contractions was expressed by the modulus of the correlation coefficient (r); in most cases,

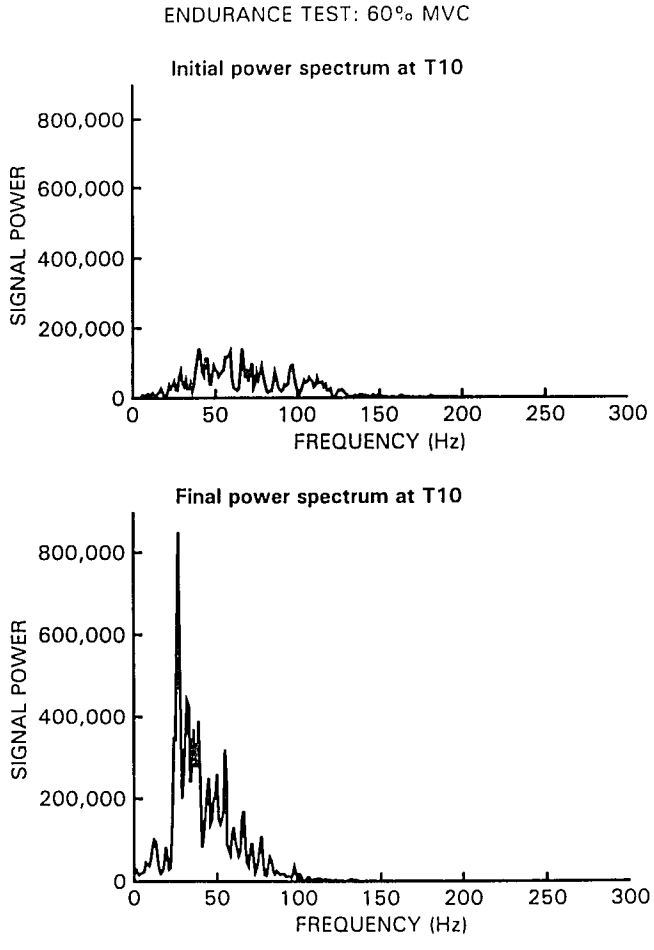


Figure 4. Initial and final power spectra recorded during the course of an isometric contraction at 60% MVC, sustained to the limit of endurance. At the end of the contraction, power at low frequencies is increased markedly, and at high frequencies, is reduced slightly.

Table 2. Changes in Spectral Parameters of the Surface EMG at the Level of the 10th Thoracic (T) and Third Lumbar (L) Vertebrae

		Amplitude Spectrum			Power Spectrum		
	Site	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)
Median frequency	T	−10 (13)	−25 (13)	−20 (11)	−11 (11)	−24 (15)	−17 (13)
	L	−10 (10)	−27 (16)	−13 (8)	−9 (8)	−23 (16)	−13 (4)
Total power	T	32 (29)	22 (28)	20 (30)	110 (137)	190 (304)	98 (113)
	L	30 (30)	37 (48)	28 (17)	128 (115)	142 (106)	111 (102)
Peak amplitude	T	50 (55)	73 (99)	41 (43)	150 (232)	146 (153)	128 (143)
	L	49 (39)	96 (39)	41 (41)	183 (259)	249 (189)	136 (144)
Peak amplitude frequency	T	0 (37)	−12 (42)	−13 (27)	0 (37)	−12 (42)	−13 (27)
	L	2 (35)	−16 (31)	−17 (15)	2 (35)	−16 (31)	−17 (15)

Values shown are the mean (standard deviation) percentage change from initial values during three loading regimes: 1) 60% maximum voluntary contraction (MVC) for 46 seconds (60% submax), 2) 60% MVC to the limit of endurance (60% LOE), and 3) 80% MVC for 23 seconds (80% submax). Data were calculated from amplitude and power spectra.

r was between 0.5–0.7, but lower values were obtained for the peak amplitude frequency (Table 3). In the case of the median frequency, r was considerably higher for the amplitude spectra than the power spectra.

Frequency Bands

Changes in the ten frequency bands of the amplitude and power spectra during the three tests are compared in Figures 5A and B respectively. During the 60% submaximal test, the EMG signal increased slightly at high frequencies and markedly at low frequencies. However, the 60% endurance test caused the high frequency signal to decrease, and low frequency signal to increase dramatically, especially in the two lowest bands (5–30 Hz and 30–60 Hz). Changes in the 80% submaximal test were intermediate between the other two tests. The linearity of the change in each band is given by the modulus of the correlation coefficient r in Table 4. This was highest in the lowest frequency band, where it was between 0.65 and 0.81.

Predictors of Endurance Time

The mean time to endurance in the 60% (endurance) test was 91 sec (SD 43 sec, range: 40–170 sec). The gradients of the linear regression lines describing the change of each spectral parameter with respect to time showed no correlation with endurance time. This was true for both the amplitude and power spectrum data, and for both the 60% submaximal test and the 60% endurance test. However, the rate of increase in the signal in the first frequency band at T10 in the 60% submaximal test was a significant predictor of endurance time when the test was continued to the endurance limit (r = 0.52).

Repeatability of the Measurements

The intraclass correlation coefficients for within-day and day-to-day repeated trials are shown for the amplitude spectrum parameters in Table 5. With the exception of the peak amplitude frequency, initial values for all parameters, indicated by the intercept of the linear

Table 3. Mean Correlation Coefficients (r) for the Changes in Spectral Parameters With Respect to Time During the Three Loading Regimes

		Amplitude Spectrum			Power Spectrum		
	Site	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)
Median frequency	T	0.63	0.82	0.81	0.47	0.61	0.60
	L	0.59	0.73	0.65	0.38	0.56	0.52
Total power	T	0.66	0.50	0.55	0.70	0.64	0.57
	L	0.66	0.57	0.55	0.68	0.62	0.53
Peak amplitude	T	0.55	0.60	0.49	0.55	0.57	0.49
	L	0.61	0.62	0.50	0.58	0.61	0.50
Peak amplitude frequency	T	0.31	0.37	0.41	0.31	0.37	0.41
	L	0.31	0.34	0.45	0.31	0.34	0.45

See Table 2 for definition of loading regimes. LOE = limit of endurance. T = thoracic. L = lumbar.

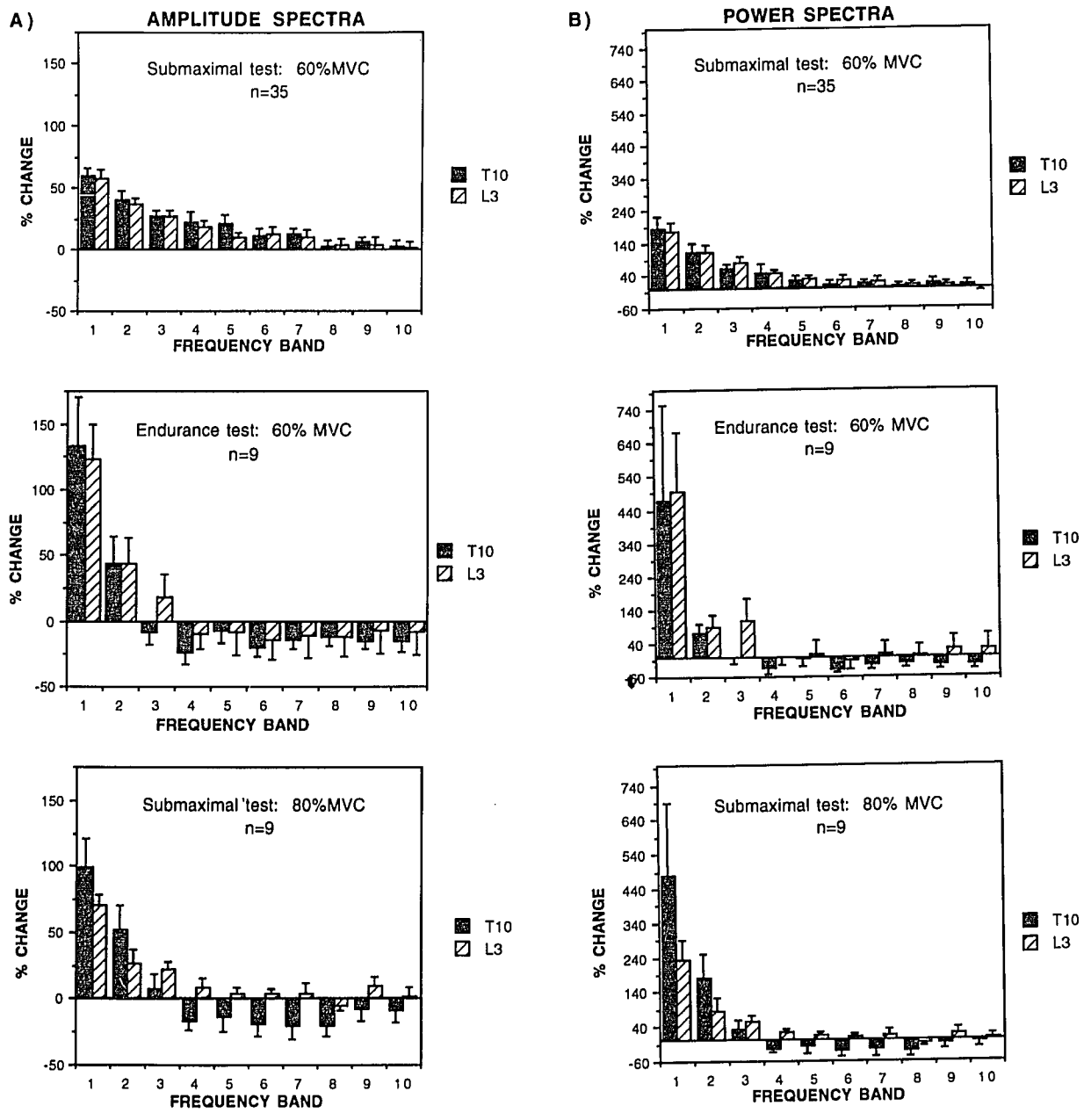


Figure 5. The percentage change in the EMG signal during the sustained isometric contractions is shown for each of ten discrete frequency bands, ranging from band 1 (5–30 Hz) up to band 10 (270–300 Hz). Results are compared for the three different fatigue tests and are calculated separately from the amplitude spectra (A) and the power spectra (B).

regression line, showed good reproducibility ($R \geq 0.76$ in all cases), and there was little difference between the within-day and day-to-day values. However, the changes in the parameters with respect to time (the gradients) were more variable. In this case, the increases in signal in the first two frequency bands and in the total power proved to be the most reproducible changes in the within-day trials. In the day-to-day trials, only the signal in the first two bands achieved R values that were always above 0.8. The repeatability of power spectrum parameters is not shown in Table 5; in most cases, the R values were slightly lower than for the amplitude spectrum.

Discussion

Surface EMG signals are inherently "noisy," and the effects of noise will be considered before the results are discussed. Reducing skin resistance below 10 kohms, and often below 2 kohms, increased the signal-to-noise ratio to the extent that rapid flicking of the EMG leads generated a signal less than 10% of that obtained from an MVC. During the static contractions in this study, the artifact would be much less than this. Movement artifacts occur at low frequencies, and for this reason, the high-pass filter was set at 5 Hz, and frequencies below this were excluded from the analysis. It has been

Table 4. Mean Correlation Coefficients (r) for the Change in EMG Signal in Each Frequency Band With Respect to Time During the Three Loading Regimes

Frequency Band (Hz)	Site	Amplitude Spectrum			Power Spectrum		
		60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)	60% submax (n = 35)	60% LOE (n = 9)	80% submax (n = 9)
5–30	T	0.70	0.81	0.80	0.67	0.75	0.75
	L	0.68	0.74	0.75	0.65	0.70	0.70
30–60	T	0.63	0.55	0.59	0.63	0.55	0.60
	L	0.58	0.44	0.43	0.59	0.46	0.45
60–90	T	0.52	0.46	0.52	0.51	0.43	0.50
	L	0.54	0.57	0.53	0.54	0.55	0.53
90–120	T	0.49	0.54	0.46	0.49	0.51	0.41
	L	0.47	0.49	0.41	0.46	0.44	0.43
120–150	T	0.45	0.43	0.65	0.44	0.44	0.66
	L	0.39	0.59	0.23	0.40	0.56	0.27
150–180	T	0.43	0.49	0.69	0.43	0.51	0.67
	L	0.40	0.58	0.24	0.41	0.58	0.27
180–210	T	0.44	0.47	0.57	0.43	0.47	0.56
	L	0.45	0.64	0.31	0.45	0.63	0.34
210–240	T	0.46	0.39	0.60	0.46	0.40	0.60
	L	0.46	0.59	0.28	0.46	0.59	0.27
240–270	T	0.45	0.36	0.44	0.45	0.34	0.43
	L	0.45	0.61	0.35	0.46	0.61	0.38
270–300	T	0.46	0.37	0.46	0.46	0.37	0.44
	L	0.35	0.61	0.26	0.37	0.61	0.25

See Table 2 for definition of loading regimes. LOE = limit of endurance. T = thoracic. L = lumbar.

suggested that all frequencies below 20 Hz be excluded if movement artifact is to be eliminated entirely, but this would be unacceptable for the erector spinae, which generate a substantial signal at low frequencies (Figure 4). The signal in the frequency band 5–30 Hz is unlikely to include much noise because it was the most repeatable of all the parameters studied (Table 5) and also showed the most linear changes with time (Table 4). We have found previously that our T10 electrodes pick up an ECG artifact, most of it in the frequency range of 5–30 Hz. We assessed the size of the ECG artifact in five subjects by positioning each of them in the test position (70% flexion) but with their upper body supported so that the erector spinae was silent. The remaining signal could then be attributed to the ECG rather than muscle activity. The ECG signal was recorded in this manner before and after the isometric contractions and was then analyzed using fast fourier transforms to determine its size in each of the frequency bands. The ECG artifact accounted for less than 10% of the initial signal in the 5–30 Hz band and altered little during fatiguing contractions. Hence, changes in the lowest frequency band appear to reflect real fatigue-induced changes in the erector spinae muscles. A high frequency cut-off of 300 Hz was used because the signal above this was small and varied little during the contractions. Other investigators have reported few signals above 300 Hz in the erector spinae.[27] This may reflect the predominance of slow

Table 5. Intraclass Correlation Coefficients (r) for Initial Values of the Spectral Parameters (intercept, I) and Their Rate of Change With Respect to Time (gradient, G) when the 60% Submaximal Test Was Repeated on the Same Day After 1 Hour (n = 10) and on Separate Days (n = 6)

	Within Day				Day to Day			
	T10		L3		T10		L3	
	I	G	I	G	I	G	I	G
Median frequency	0.94	0.92	0.97	0.50	0.90	0.70	0.85	−3.4
Total power	0.99	0.93	0.92	0.99	0.94	0.50	0.88	0.84
Peak amplitude	0.97	0.88	0.76	0.68	0.94	0.73	0.92	0.84
Peak amplitude frequency	0.37	0.05	−0.27	−1.2	−2.89	−6.90	0.37	0.31
5–30 Hz	0.90	0.92	0.93	0.96	0.98	0.90	0.97	0.91
30–60 Hz	0.98	0.86	0.89	0.92	0.94	0.82	0.91	0.93

Values of G and I were calculated by linear regression.

twitch (Type I) fibers in these muscles, as well as the small size of the fast twitch (Type II) fibers, which have slower conduction velocities than the larger Type II fibers found in other muscles.[11]

Our subjects were tested in 70% lumbar flexion because the lumbar spine is flexed by 70–100% during everyday bending and lifting activities, even when the knees are bent.[1,18,19] The low end of this range (70% flexion) was chosen to minimize tension in the intervertebral ligaments and lumbodorsal fascia and ensure that most of the extensor moment would be generated by the erector spinae muscles, both at T10 and L3.[19] However, EMG changes in some subjects could be much more marked at one level than the other, even in 70% flexion, suggesting that passive tissues might become involved at a local level. This highlights the importance of recording at more than one spinal level when assessing fatigue in back muscles.

Most investigators have studied subjects in lordotic postures, where the erector spinae are in a shortened position compared with the lengthened position adopted during lifting. Lengthened muscles have a reduced EMG signal per unit force[3,34] and exhibit a shift in the power spectrum to lower frequencies compared with when they are shortened.[36] This shift cannot be explained fully by an increase in the interelectrode distance[6,36] and must be due predominantly to some other factor. These findings may explain why our measurements of initial median frequency of the power spectrum (39–56 Hz: Table 1) were below those reported previously in lordotic postures.[7,29,36,37]

Despite this dependence of the EMG signal on muscle length, comparisons can be drawn with previous work. In this experiment, increasing the force of contraction from 60% to 80% MVC had little effect on the initial median frequency, whereas several studies on the erector spinae have shown median frequency to decrease with increasing force of contraction.[37,39] On the other hand, previous studies on limb muscles have shown the opposite trend[9,12] or found no dependence on load.[6,9] These disparate results may reflect different methods by which muscles increase their force of contraction; those with small numbers of motor units, such as the first dorsal interosseus or the adductor pollicis, increase force predominantly by increasing firing rate[15,26] and this has little effect on median frequency.[42] In contrast, large muscles such as the biceps brachii and the deltoid increase their force of contraction mainly by motor unit recruitment.[15,26,40] According to the size principle,[22] the larger diameter fast twitch fibers will be recruited as the contraction force increases, and this will cause the median frequency to increase. Roy et al[37] have suggested that the opposite trend occurs in the erector spinae because in these muscles the Type II fibers that are recruited at higher force levels are smaller in diameter than the Type I fibers.[5,35,41] In the present study, the initial median frequencies were not reduced at the higher contraction force (80% MVC), but this may have been because they were reduced already, by virtue of the increased muscle length. The rate of decrease in median frequency during sustained contractions was related to contractile force in the present and previous experiments,[37] and this is a consistent finding provided that the work load is above approximately 30% MVC. The total power changes in this study were also dependent on load. Both the initial values and the rate of increase were significantly higher in the 80% submaximal test compared with the 60% submaximal test, in agreement with earlier observations of the RMS and integrated EMG signal.[8,12,36,43] The lack of correlation between the rate of change of the spectral parameters and endurance time in this study may also be because of factors associated with the muscle length. The already considerable compression of the spectrum to low frequencies at the start of the test may cause the fatigue-induced shift to be less marked. The low percentage decrease in median frequency at the endurance limit seems to support this suggestion. We also used a high relative load at a muscle length where the MVC would also have been high compared with erect postures, so the ability of the subjects to reach their true endurance limit may have been more dependent on motivation in this particular test.

The detailed "banding" analysis carried out in this study enabled us to identify repeatable and linear EMG indices of the changes that occur in fatiguing muscles. In previous attempts to band the power spectrum, two or three frequency bands were chosen to represent the high and low frequency components of the signal,[3,8,33] and the high/low frequency ratio reflected fatigue-induced changes in the muscle. However, other studies found that arbitrarily chosen frequency ratios were less reliable than the median frequency as an index of fatigue.[21,38] In the present study, the signal in the frequency range 5–30 Hz showed the most consistent changes with fatigue at all loads and all duration times and also demonstrated the most linear change with respect to time (Table 4). As a result, the rate of increase in the signal in this frequency band during the 60% submaximal test was the only parameter that was a significant predictor of endurance time in those subjects who continued to their endurance limit. The reason why the gradient in the endurance test was not similarly correlated to endurance time is not clear. Subtle changes in the gradient as the endurance limit was approached may have been responsible, in which case the selection of an appropriate contraction time for submaximal testing is something that would have to be considered in future investigations. Median frequency, peak amplitude, and total power also changed significantly during each of the three fatigue tests, but the changes were smaller and less consistent than those of the signal in the 5–30 Hz frequency band, and they were less repeatable on a day-to-day basis. The linearity of the decrease in median frequency

with contraction time was improved if the amplitude rather than the power spectrum data were considered (Table 3). This may be because the power spectrum of the erector spinae is dominated by the high amplitude signals at low-to-middle frequencies, whereas the amplitude spectrum gives more prominence to small signals at the lowest and highest frequencies. Hence, the power spectrum will be less sensitive than the amplitude spectrum to fatigue-induced shifts from high to low frequencies.

The detailed frequency banding of this study explains the previously observed changes in median frequency and total power during sustained fatiguing contractions. During all tests, the low frequency signal increased throughout the contraction (Figure 5A, B), and this no doubt explains why the median frequency decreases. The middle-to-high frequency signal appeared to show a biphasic response: an initial increase, followed by a subsequent decrease as the endurance limit was approached (Figure 5A, B). The initial increase is evidently too slight to overcome the increase at low frequencies and cause the median frequency to increase, but together with the low-frequency increase, it would contribute to a marked overall increase in total power. However, if the endurance limit was approached, the diminishing signal at middle-to-high frequencies would counteract the continuing increase at low frequencies so that total power could increase more slowly, level off, or even decrease. Such an effect was observed in some individuals in the present study, and it could explain why changes in the RMS and integrated EMG during contractions maintained to the endurance limit are sometimes greater during the early part of the contraction.[14,39]

Increases in the low frequency EMG signal during a sustained isometric contraction have been attributed to a slowing of the conduction velocity in the active fibers and an increased duration of the motor unit action potential.[25,28] Increased synchronization of motor unit action potentials has also been proposed as a contributing factor.[8,43] As far as the authors are aware, increases in the high frequency EMG signal have not been reported previously; they suggest that some recruitment of additional motor units occurs during the initial stages of a sustained submaximal isometric contraction. An alternative explanation, that some fibers increase their firing rate, is unlikely to be applicable to a large muscle group like the erector spinae, for the reasons discussed earlier. Recruitment of additional motor units may be accompanied by a slowing of conduction velocity and/or firing rate in those fibers that are already active, which would contribute to the increased signal observed in the low frequency bands. As the contraction continues, the latter effect probably will predominate, and this would explain why the signal in the high frequency bands eventually decreases (Figure 5A, B).

Comparing the EMG changes in response to different loading conditions gives some clues about the metabolic processes involved in fatigue. Several investigators have reported little or no change in the amplitude of the surface EMG signal[13,39] during isometric contractions less than 30% MVC. Under these conditions blood flow to the muscle may be increased[32] and accumulation of metabolic end-products will be minimal. At higher levels of force, isometric contractions increase the amplitude of the surface EMG[14,43,44] and reduce its median frequency.[25,31,36,37] They may also occlude blood flow to the muscle[23,32] causing ischemia to develop, lactic acid to accumulate, and muscle pH to decrease. Whether these changes become limiting will depend on the level of contraction because this determines the amount of occlusion as well as the rate of substrate metabolism. The lack of EMG changes at loads less than 30% MVC suggest that, under these conditions, contractions may be limited by central factors such as motivation or pain; at higher loads, changing metabolic conditions within the muscle probably limit performance. The present results suggest that during submaximal contractions of the erector spinae, the metabolic effects of fatigue may be counteracted or delayed by the recruitment of additional motor units. The beneficial effects of this are likely to be greatest during the initial stages of low to moderate contractions when the potential for recruitment will be at its highest. Such a compensatory mechanism has not been demonstrated before, and it may reflect a functional requirement of the back muscles to maintain static "postural" contractions for long periods.

Conclusions

1. The surface EMG signal offers reliable indices of fatigue-induced changes in the erector spinae muscles during sustained isometric contractions.
2. Of these indices, the signal in the frequency band 5–30 Hz showed the most consistent, repeatable and linear changes with time.
3. Changes can vary with electrode location, and both thoracic and lumbar portions of the erector spinae should be monitored.
4. Increases in the middle-to-high frequency component of the EMG signal suggest that motor unit recruitment may occur during the early stages of a sustained submaximal isometric contraction. This may be a compensatory mechanism to delay the onset of fatigue.

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